Solutions to Axler's Linear Algebra

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1 Vector Spaces

1A $\mathbb{R}^n$ and $\mathbb{C}^n$

Exercise 1

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$ and $\beta = c + di$ for some $a, b, c, d \in \mathbb{R}$. Then

$$\begin{aligned}\alpha + \beta &= (a + bi) + (c + di)\\ &= (a + c) + (b + d)i\\ &= (c + a) + (d + b)i\\ &= (c + di) + (a + bi) = \beta + \alpha.\end{aligned}$$

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Exercise 2

Show that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha + \beta) + \gamma &= ((a + bi) + (c + di)) + (e + fi)\\ &= ((a + c) + (b + d)i) + (e + fi)\\ &= (a + c + e) + (b + d + f)i\\ &= (a + (c + e)) + (b + (d + f))i\\ &= (a + bi) + ((c + di) + (e + fi)) = \alpha + (\beta + \gamma).\end{aligned}$$

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Exercise 3

Show that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha\beta)\gamma &= ((a + bi)(c + di))(e + fi)\\ &= ((ac - bd) + (ad + bc)i)(e + fi)\\ &= ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i\\ &= (a(ce) - b(de) - a(df) - b(cf)) + (a(cf) - b(df) + a(de) + b(ce))i\\ &= \alpha(\beta\gamma).\end{aligned}$$

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Exercise 4

Show that $\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}\gamma(\alpha + \beta) &= (e + fi)((a + bi) + (c + di))\\ &= (e + fi)((a + c) + (b + d)i)\\ &= (e(a + c) - f(b + d)) + (e(b + d) + f(a + c))i\\ &= (ea - fb + ec - fd) + (eb + fa + ed + fc)i\\ &= (ea - fb) + (eb + fa)i + (ec - fd) + (ed + fc)i\\ &= (e + fi)(a + bi) + (e + fi)(c + di) = \gamma\alpha + \gamma\beta.\end{aligned}$$

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Exercise 5

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$. We want to find $\beta = c + di$ such that $\alpha + \beta = 0$. This gives

$$(a + bi) + (c + di) = (a + c) + (b + d)i = 0 + 0i.$$

Solving for c and d, we get $c = -a$ and $d = -b$. Hence, $\beta = c + di = -a - bi = -\alpha$.

To see that β is unique, suppose there is another $\gamma \in \mathbb{C}$ such that $\alpha + \gamma = 0$. Then

$$\beta = \beta + 0 = \beta + (\alpha + \gamma) = (\beta + \alpha) + \gamma = 0 + \gamma = \gamma.$$

Hence, β is unique. ∎

Exercise 6

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$ with $\alpha \neq 0$. We want to find $\beta = c + di$ such that $\alpha\beta = 1$. This gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i = 1 + 0i.$$

Solving for c and d, we get the system of equations

$$\begin{cases} ac - bd = 1 \\ ad + bc = 0 \end{cases} \implies \begin{cases} a^2c - abd = a \\ abd + b^2c = 0 \end{cases}$$

Adding the two equations, we get

$$(a^2 + b^2)c = a \implies c = \frac{a}{a^2 + b^2}.$$

Substituting this into the second equation, we get

$$abd + b^2\frac{a}{a^2 + b^2} = 0 \implies d = \frac{-b}{a^2 + b^2}.$$

Hence,

$$\beta = c + di = \frac{a}{a^2 + b^2} + \frac{-b}{a^2 + b^2}i.$$

If $\gamma \in \mathbb{C}$ is another complex number such that $\alpha\gamma = 1$, then

$$\gamma = \gamma 1 = \gamma(\alpha\beta) = (\gamma\alpha)\beta = 1\beta = \beta.$$

Hence, β is unique. ∎

Exercise 7

Show that

$$\frac{-1 + \sqrt{3}i}{2}$$

is a cube root of 1.

Solution. Note that the complex number given can be written as

$$\frac{1}{2}(-1 + \sqrt{3}i).$$

Cubing $-1+\sqrt{3}i$, we get

$$\begin{aligned}(-1+\sqrt{3}i)^3 &= (-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3 \\ &= -1 + 3\sqrt{3}i + 3(-1)(-3) + (\sqrt{3}i)^3 \\ &= -1 + 3\sqrt{3}i + 9 + 3\sqrt{3}i(-i) \\ &= -1 + 3\sqrt{3}i + 9 - 3\sqrt{3}i \\ &= 8.\end{aligned}$$

Therefore,

$$\left(\frac{-1+\sqrt{3}i}{2}\right)^3 = \frac{(-1+\sqrt{3}i)^3}{8} = \frac{8}{8} = 1. \qquad \blacksquare$$

Exercise 8

Find two distinct square roots of i.

Solution. Let $z = a + bi$ for some $a, b \in \mathbb{R}$. We want to find z such that $z^2 = i$. This gives

$$(a+bi)^2 = a^2 - b^2 + 2abi = 0 + 1i.$$

Solving for a and b, we get the system of equations

$$\begin{cases} a^2 - b^2 = 0 \\ 2ab = 1 \end{cases}$$

The first equation yields $a = b$ or $a = -b$. If $a = b$, then the second equation gives

$$2b^2 = 1 \implies b = \pm\frac{1}{\sqrt{2}} \implies a = \pm\frac{1}{\sqrt{2}}.$$

If $a = -b$, this results in $-2b^2 = 1$ which has no solution when $b \in \mathbb{R}$. Hence, the two distinct square roots of i are

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \quad \text{and} \quad -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i. \qquad \blacksquare$$

Exercise 9

Find $x \in \mathbb{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

Solution. Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. The equation becomes

$$(4, -3, 1, 7) + 2(x_1, x_2, x_3, x_4) = (5, 9, -6, 8),$$

which simplifies to the system of equations

$$\begin{cases} 5 = 4 + 2x_1 \\ 9 = -3 + 2x_2 \\ -6 = 1 + 2x_3 \\ 8 = 7 + 2x_4 \end{cases} \implies \begin{cases} 2x_1 = 1 \\ 2x_2 = 12 \\ 2x_3 = -7 \\ 2x_4 = 1 \end{cases}$$

Therefore,

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right). \qquad \blacksquare$$

Exercise 10

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2-3i, 5+4i, -6+7i) = (12-5i, 7+22i, -32-9i).$$

Solution. Suppose there exists $\lambda \in \mathbb{C}$ such that

$$\lambda(2-3i, 5+4i, -6+7i) = (12-5i, 7+22i, -32-9i).$$

This gives the system of equations

$$\begin{cases} \lambda(2-3i) = 12-5i \\ \lambda(5+4i) = 7+22i \\ \lambda(-6+7i) = -32-9i \end{cases}$$

Solving the first equation for λ, we get

$$\lambda = \frac{12-5i}{2-3i} = \frac{(12-5i)(2+3i)}{(2-3i)(2+3i)} = \frac{24+36i-10i-15i^2}{4+9} = \frac{24+26i+15}{13} = 3+2i.$$

Solving the second equation for λ, we get

$$\lambda = \frac{7+22i}{5+4i} = \frac{(7+22i)(5-4i)}{(5+4i)(5-4i)} = \frac{35-28i+110i-88i^2}{25+16} = \frac{35+82i+88}{41} = 3+2i.$$

Solving the third equation for λ, we get

$$\lambda = \frac{-32-9i}{-6+7i} = \frac{(-32-9i)(-6-7i)}{(-6+7i)(-6-7i)} = \frac{192+224i+54i+63i^2}{36+49} = \frac{192+278i-63}{85} = \frac{129+278i}{85}.$$

Since the value of λ obtained from the third equation is different from the value obtained from the first two equations, no such $\lambda \in \mathbb{C}$ exists. ■

Exercise 11

Show that $(x+y)+z = x+(y+z)$ for all $x, y, z \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, and $z = (z_1, z_2, \ldots, z_n)$. Then

$$\begin{aligned} (x+y)+z &= ((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) + (z_1, z_2, \ldots, z_n) \\ &= ((x_1+y_1), (x_2+y_2), \ldots, (x_n+y_n)) + (z_1, z_2, \ldots, z_n) \\ &= (x_1+y_1+z_1, x_2+y_2+z_2, \ldots, x_n+y_n+z_n) \\ &= (x_1, x_2, \ldots, x_n) + ((y_1+z_1), (y_2+z_2), \ldots, (y_n+z_n)) \\ &= x+(y+z). \end{aligned}$$

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Exercise 12

Show that $(ab)x = a(bx)$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned} (ab)x &= (ab)(x_1, x_2, \ldots, x_n) \\ &= ((ab)x_1, (ab)x_2, \ldots, (ab)x_n) \\ &= (a(bx_1), a(bx_2), \ldots, a(bx_n)) \\ &= a((bx_1, bx_2, \ldots, bx_n)) \\ &= a(bx). \end{aligned}$$

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Exercise 13

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$. Then

$$\begin{aligned} 1x &= 1(x_1, x_2, \ldots, x_n) \\ &= (1x_1, 1x_2, \ldots, 1x_n) \\ &= (x_1, x_2, \ldots, x_n) \\ &= x. \end{aligned}$$

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Exercise 14

Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ be elements of $\mathbb{F}^n$ and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} \lambda(x + y) &= \lambda((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) \\ &= \lambda((x_1 + y_1), (x_2 + y_2), \ldots, (x_n + y_n)) \\ &= (\lambda(x_1 + y_1), \lambda(x_2 + y_2), \ldots, \lambda(x_n + y_n)) \\ &= (\lambda x_1 + \lambda y_1, \lambda x_2 + \lambda y_2, \ldots, \lambda x_n + \lambda y_n) \\ &= (\lambda x_1, \lambda x_2, \ldots, \lambda x_n) + (\lambda y_1, \lambda y_2, \ldots, \lambda y_n) \\ &= \lambda(x_1, x_2, \ldots, x_n) + \lambda(y_1, y_2, \ldots, y_n) \\ &= \lambda x + \lambda y. \end{aligned}$$

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Exercise 15

Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned} (a + b)x &= (a + b)(x_1, x_2, \ldots, x_n) \\ &= ((a + b)x_1, (a + b)x_2, \ldots, (a + b)x_n) \\ &= (ax_1 + bx_1, ax_2 + bx_2, \ldots, ax_n + bx_n) \\ &= (ax_1, ax_2, \ldots, ax_n) + (bx_1, bx_2, \ldots, bx_n) \\ &= ax + bx. \end{aligned}$$

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